Assignment - 3

Mayur Zope TE Comp A 71

Tic Tac Toe AI using Minmax with Alpha Beta Pruning.

```
import math
# Board
board = [" " for _ in range(9)]

def print_board():
    print()
    for i in range(3):
        print(board[i*3] + " | " + board[i*3+1] + " | " + board[i*3+2])
        if i < 2:
            print("--+---+--")
    print()

def check_winner(player):
    win_positions = [
        [0,1,2],[3,4,5],[6,7,8],
        [0,3,6],[1,4,7],[2,5,8],
        [0,4,8],[2,4,6]
    ]
    for pos in win_positions:
        if board[pos[0]] == board[pos[1]] == board[pos[2]] == player:
            return True
    return False

def is_draw():
    return " " not in board

def minimax(is_maximizing, alpha, beta):
    if check_winner("O"):
        return 1
    if check_winner("X"):
```

```
        return -1
    if is_draw():
        return 0

    if is_maximizing:
        best_score = -math.inf
        for i in range(9):
            if board[i] == " ":
                board[i] = "O"
                score = minimax(False, alpha, beta)
                board[i] = " "
                best_score = max(score, best_score)
                alpha = max(alpha, score)
                if beta <= alpha:
                    break
        return best_score
    else:
        best_score = math.inf
        for i in range(9):
            if board[i] == " ":
                board[i] = "X"
                score = minimax(True, alpha, beta)
                board[i] = " "
                best_score = min(score, best_score)
                beta = min(beta, score)
                if beta <= alpha:
                    break
        return best_score

def ai_move():
    best_score = -math.inf
    move = -1
```

```
    for i in range(9):
        if board[i] == " ":
            board[i] = "O"
            score = minimax(False, -math.inf, math.inf)
            board[i] = " "
            if score > best_score:
                best_score = score
                move = i

    board[move] = "O"

def player_move():
    move = int(input("Enter position (1-9): ")) - 1
    if board[move] == " ":
        board[move] = "X"
    else:
        print("Invalid move! Try again.")
        player_move()

def play_game():
    print("Tic-Tac-Toe (You = X, AI = O)")
    print_board()

    while True:
        player_move()
        print_board()

        if check_winner("X"):
            print("You Win!")
            break
        if is_draw():
```

```
            print("Draw!")
            break

        print("AI is thinking...")
        ai_move()
        print_board()

        if check_winner("O"):
            print("AI Wins!")
            break
        if is_draw():
            print("Draw!")
            break

play_game()
```

// OUTPUT

```
hardwarelab00@hardwarelab00-ThinkCentre-M70s:~$ python3 assignment3.py
Tic-Tac-Toe (You = X, AI = O)

  |   |
--+---+--
  |   |
--+---+--
  |   |

Enter position (1-9): 5

  |   |
--+---+--
  | X |
--+---+--
  |   |

AI is thinking...

O |   |
--+---+--
  | X |
--+---+--
  |   |

Enter position (1-9): 7

O |   |
--+---+--
  | X |
--+---+--
X |   |

AI is thinking...
```

```
AI is thinking...

O | X | O
--+---+--
  | X |
--+---+--
X | O |

Enter position (1-9): 4

O | X | O
--+---+--
X | X |
--+---+--
X | O |

AI is thinking...

O | X | O
--+---+--
X | X | O
--+---+--
X | O |

Enter position (1-9): 9

O | X | O
--+---+--
X | X | O
--+---+--
X | O | X

Draw!
```